Analyzing Client Satisfaction and Service Utilization at the Dempsey Center

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I. Introduction

The Dempsey Center is a nonprofit that provides free support services to people affected by cancer in Maine. Since opening, they've offered counseling, support groups, complementary therapies like massage and acupuncture, movement programs, and educational workshops. The goal is to help people deal with the physical, emotional, and social challenges that come with cancer.

Every year, Dempsey runs a client survey to see how satisfied people are, what's working, and what needs improvement. We analyzed four years of these surveys, from 2022 through 2025, which yielded more than 1,100 responses for analysis. We focused on three main questions:

1. Has client satisfaction gotten better over these four years?
2. What factors actually predict whether someone will be highly satisfied?

The purpose of this analysis is to provide the Dempsey Center with data-driven insights that can inform program improvement and resource allocation. If we can figure out what makes clients happy and how different groups engage with services, Dempsey can make better decisions about where to put their resources and how to improve their programs.

II. Background

Cancer affects more than just physical health. A 2024 review in eClinicalMedicine found that cancer patients worldwide face significant unmet psychosocial needs, including emotional distress and social isolation, which directly impact their quality of life and health outcomes (Gao et al., 2024). However, access to support services remains inconsistent, and many patients who could benefit from these services are unaware of them.

A 2021 scoping review in PLOS ONE evaluated psychosocial cancer support programs and found that these services can improve well-being and successfully address distressing concerns for both patients and their families (Sautier et al., 2021). The review also emphasized that measuring client satisfaction is important for understanding service quality and identifying where improvements are needed.

The Dempsey Center's approach addresses cancer holistically. Instead of focusing only on medical treatment, they offer emotional support through counseling and groups, physical support through

complementary therapies and movement programs, and social connection through community activities. The American Cancer Society notes that this kind of comprehensive psychosocial support can improve treatment adherence, symptom management, and overall quality of life (ACS, 2025). For organizations like Dempsey that operate on limited resources, understanding what drives client satisfaction helps them allocate those resources more effectively. This analysis examines four years of client feedback to identify patterns that Dempsey can use to improve their services.

III. Methods

A. Data Overview and Demographics

This study analyzes four years of annual client survey data from the Dempsey Center, covering 2022 through 2025. The raw datasets contained 270, 298, 293, and 319 responses respectively, with column counts ranging from 72 to 105 as the survey instrument expanded over time. Total combined responses numbered 1,179 before cleaning.

One major challenge was that column positions for equivalent questions shifted across years. For example, the satisfaction question appeared in column 0 for 2022-2024 but moved to column 4 in 2025 due to the addition of metadata fields. We manually verified each variable's location in each file before extraction to ensure we were pulling the correct data. Table 1 shows the verified column positions for key variables across all four years.

Variable	2022	2023	2024	2025
Satisfaction	0	0	0	4
Life Better	6	1	1	5
Service Frequency	27	22	34	38
Age	47	54	67	74
Gender	49	57	71	78
Cancer Status	43	50	63	68

Table 1. Variable Column Positions by Year

We extracted six core variables from each year: overall satisfaction with services (rated on a 5-point scale from Very Satisfied to Very Unsatisfied), whether services made life better (Yes/No), service utilization frequency (1 time, 2-5 times, 6-9 times, or more than 10 times), age group, gender, and current cancer status. Treatment stage and household income were also available for some years but not consistently across all four.

After cleaning, the final dataset retained 1,047 rows distributed as 257 from 2022, 262 from 2023, 252 from 2024, and 276 from 2025. This represented 89% of the original data. The demographics of this cleaned dataset are described below.

The age distribution showed that the Dempsey Center primarily serves an older adult population. The 65-74 age group made up the largest share at 385 respondents (36.8%), followed by the 55-64 group at 274 respondents (26.2%) and the 75 or older group at 197 respondents (18.8%). Combined, these three groups accounted for 82% of all respondents. Younger age groups were notably underrepresented, with only 11 respondents aged 25-34 (1.1%) and 63 aged 35-44 (6.0%).

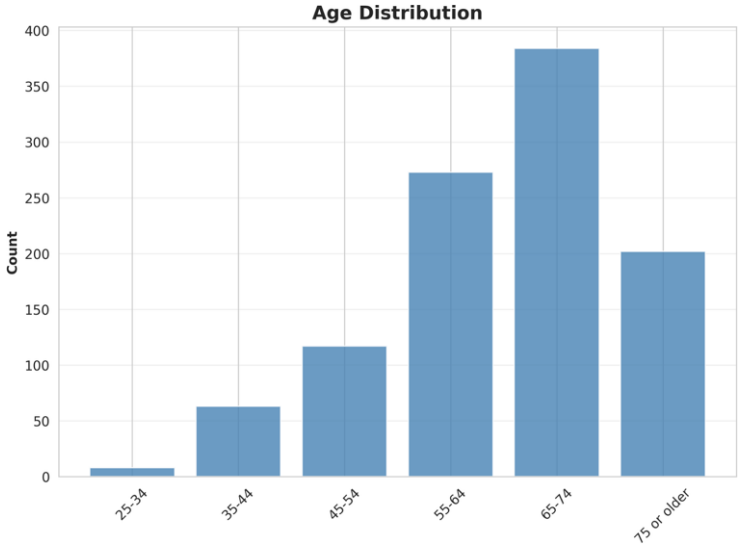

Figure 1. Age distribution of survey respondents.

Gender distribution was heavily skewed toward women, who made up 847 of 1,047 respondents (80.9%). Men accounted for 197 respondents (18.8%), and only 3 respondents (0.3%) identified as non-binary. This distribution remained stable across all four years.

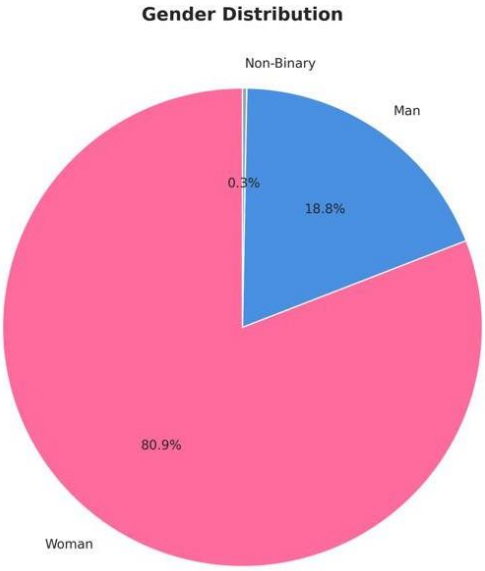

Figure 2. Gender distribution showing 81% of respondents were women.

In terms of cancer status, the largest group consisted of cancer survivors - 406 respondents (38.8%) reported "I have had cancer." The second largest group was current patients at 327 respondents

(31.2%). Approximately 17% of respondents were not patients themselves but rather care partners (10.2%) or family members and friends (6.9%) of someone with cancer. The remaining 12.9% selected "Other/prefer to self-describe."

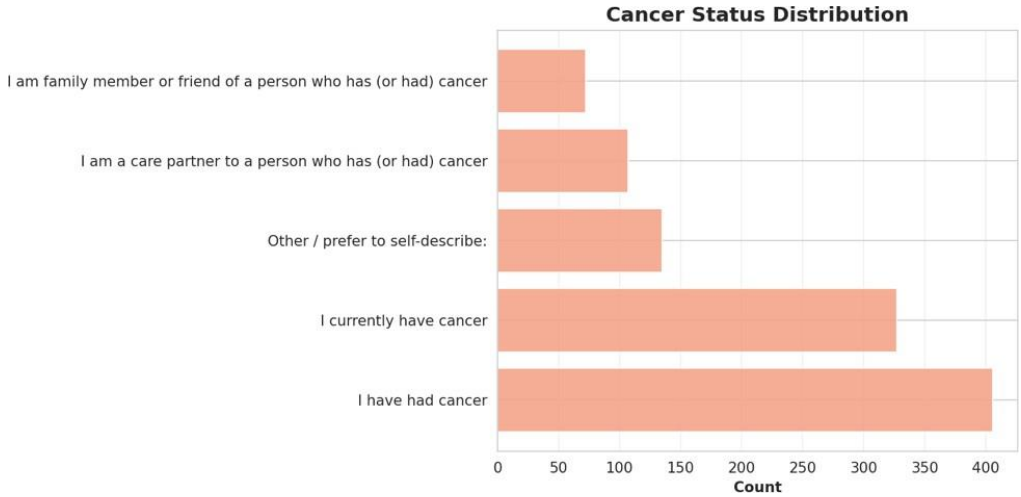

Figure 3. Cancer status distribution showing majority are current or former patients.

Service utilization patterns showed strong engagement. Among the 1,047 respondents, 465 (44.4%) reported using services more than 10 times, making this the most common category. An additional 228 (21.8%) visited 6-9 times, and 281 (26.8%) visited 2-5 times. Only 73 respondents (7.0%) had used services just once. This distribution suggests the center successfully retains clients beyond an initial visit.

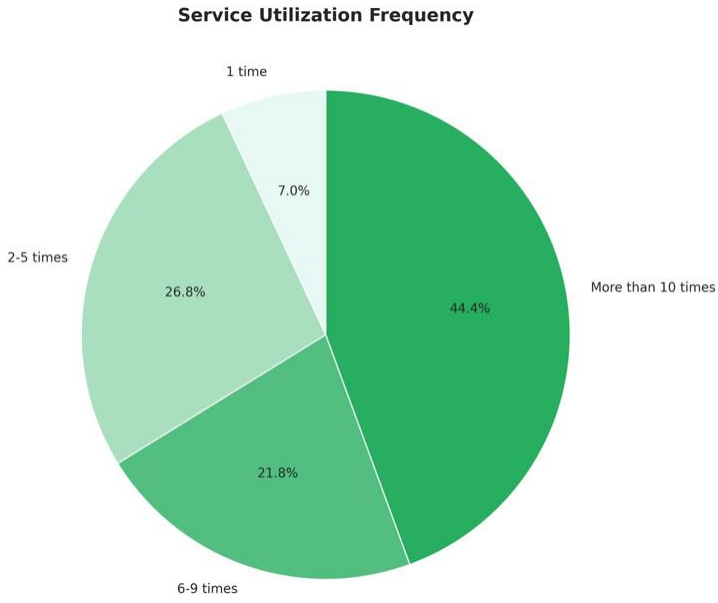

Figure 4. Service frequency distribution.

Response volume remained relatively stable across the four years. The 2025 survey had slightly higher participation (276 responses) compared to the other years, which ranged from 252 to 262 responses. This consistency suggests sustained engagement with the annual survey process.

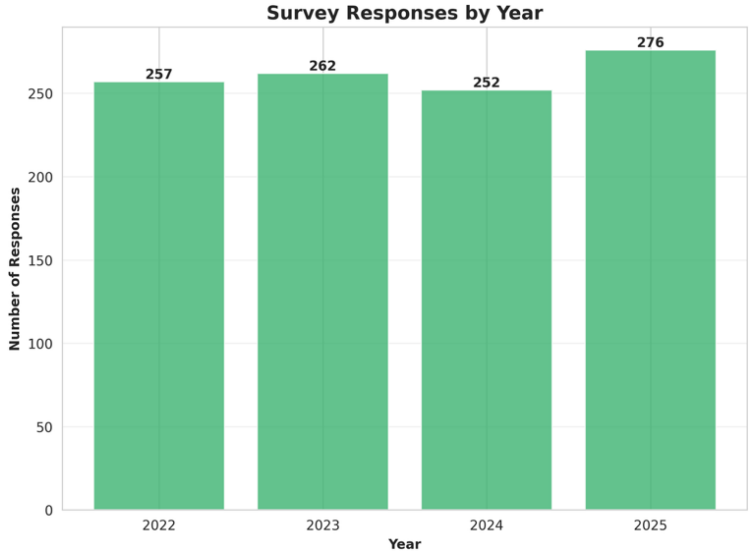

Figure 5. Number of survey responses by year.

B. Data Cleaning

Data cleaning involved several steps to ensure consistency and quality. First, we removed duplicate header rows that appeared in the data. Each Excel file contained a row with the value "Response" that was part of the column headers rather than actual data. We skipped row 0 (the true header) and row 1 (the "Response" row) when reading each file.

Second, we standardized gender labels across years. The 2022 and 2023 surveys used longer labels like "Woman/Female/Feminine" and "Man/Male/Masculine," while 2024 and 2025 used simplified labels like "Woman" and "Man." We converted all to the shorter format for consistency.

Third, we dropped rows with missing values for the two core outcome variables: Satisfaction and LifeBetter. Since our analysis centers on satisfaction, responses without this information couldn't be used. We also removed rows where respondents selected "Prefer not to answer" for Age or Gender, as these demographic variables are key predictors in our models.

The cleaning process reduced the dataset from 1,179 rows to 1,047 rows, representing an 11% reduction. The lost data was primarily from incomplete responses rather than any systematic pattern that would introduce bias.

C. Feature Encoding

Machine learning algorithms require numeric input, so we encoded all categorical variables as integers using sklearn's LabelEncoder. Age groups were encoded in ascending order: 25-34 became 0, 35-44 became 1, and so on up to 75+ becoming 5. Gender was encoded with Woman as 0, Man as 1, and Non-Binary as 2. Service frequency was encoded based on increasing usage: 1 time

became 0, 2-5 times became 1, 6-9 times became 2, and more than 10 times became 3. Cancer status was encoded alphabetically from 0 to 5.

We also created a binary target variable called HighSatisfaction for the classification task. Responses of "Very Satisfied" were coded as 1, and all other satisfaction levels (Satisfied, Neither, Unsatisfied, Very Unsatisfied) were coded as 0. This framing allowed us to specifically predict which clients would report the highest level of satisfaction.

D. Statistical and Machine Learning Methods

Chi-Square Tests of Independence. We conducted three chi-square tests to assess whether demographic and behavioral variables were associated with high satisfaction. The tests examined service frequency versus satisfaction, age versus satisfaction, and cancer status versus satisfaction. For each test, we used a significance threshold of $\alpha = 0.05$. A p-value below this threshold would indicate a statistically significant association between the variables.

Machine Learning Classification. We trained four different classifiers to predict the HighSatisfaction binary outcome using age, gender, service frequency, and cancer status as input features. The four models were Gaussian Naive Bayes (which assumes features follow normal distributions), K-Nearest Neighbors with k=3 (which classifies based on the three closest training examples), Decision Tree with maximum depth of 5 (which creates rule-based decision paths), and Random Forest with 100 estimators and maximum depth of 5 (which averages predictions from 100 decision trees). We set random_state=42 throughout to ensure reproducibility.

We split the data into 70% for training (732 samples) and 30% for testing (315 samples). After training each model on the training set, we evaluated performance on the held-out test set. We also ran 5-fold stratified cross-validation on the full dataset to get a more robust estimate of each model's performance. Stratification ensured that each fold maintained the same proportion of Very Satisfied versus other responses as the overall dataset.

For the tree-based models (Decision Tree and Random Forest), we extracted feature importance scores to understand which variables contributed most to the predictions. We also generated confusion matrices for all models to see how well they distinguished between the two classes.

Correlation Analysis. We calculated Pearson correlation coefficients between all pairs of variables, including the encoded features and the HighSatisfaction target. This helped us understand linear relationships in the data and see which features were most correlated with satisfaction.

E. Reproducibility

To support replicability, all data cleaning, analysis, and figure-generation code is available in the accompanying GitHub repository: https://github.com/Reneechang17/Dempsey-Center-Report. The analysis was conducted in Python 3.x using pandas, numpy, scikit-learn, matplotlib, and seaborn. To reproduce the results, a reader can first run the data cleaning notebook or script to generate the

cleaned dataset and then run the analysis notebook to reproduce Figures 1–21 and the reported summary tables. If raw survey files cannot be publicly shared due to privacy restrictions, the repository should include documentation describing the variable mappings, cleaning steps, and expected input file structure.

IV. Results

A. Exploratory Data Analysis

Satisfaction Trends Over Time

Overall satisfaction remained consistently high throughout the study period, though not without some variation. Satisfaction rates ranged from 75.1% in 2022 to 79.4% in 2024, with no consistent upward or downward trend overall. While the distribution of other response categories showed minor fluctuations year to year, the overall picture suggests stable service quality across all four years.

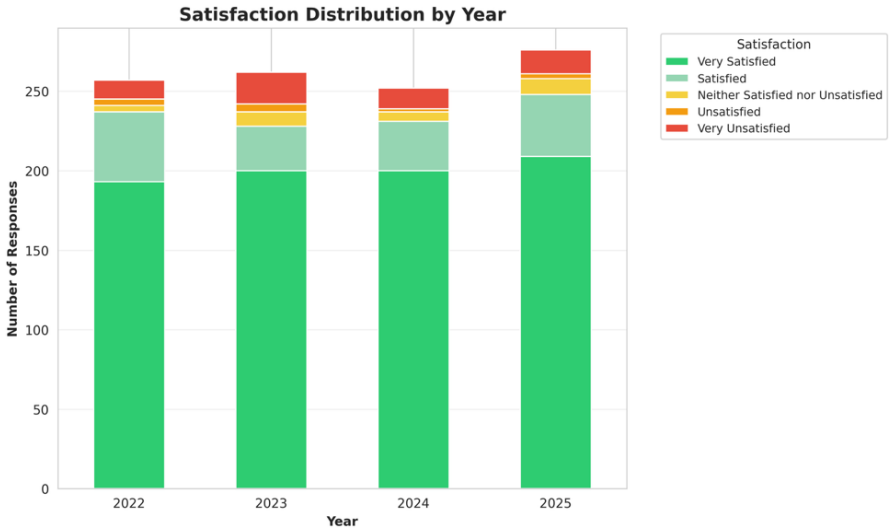

Figure 6. Satisfaction ratings by year.

Between 95% and 97% of respondents each year reported that Dempsey Center services made their life better. This was the most consistently positive finding across the entire dataset, with virtually no meaningful variation across years.

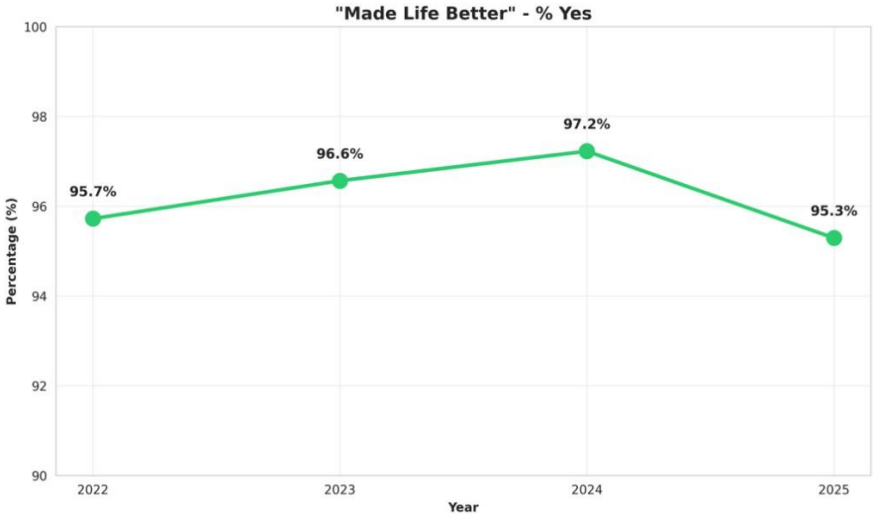

Figure 7. Proportion of respondents who reported services made their life better.

Emotional wellbeing tops the list of benefits, with over 60% of clients reporting reduced distress through the center's support. Nearly half also valued the sense of community and social connection fostered by their programs. Notably, while physical symptom relief ranks lower at 34%, the psychological and social impacts appear to be the center's strongest value drivers.

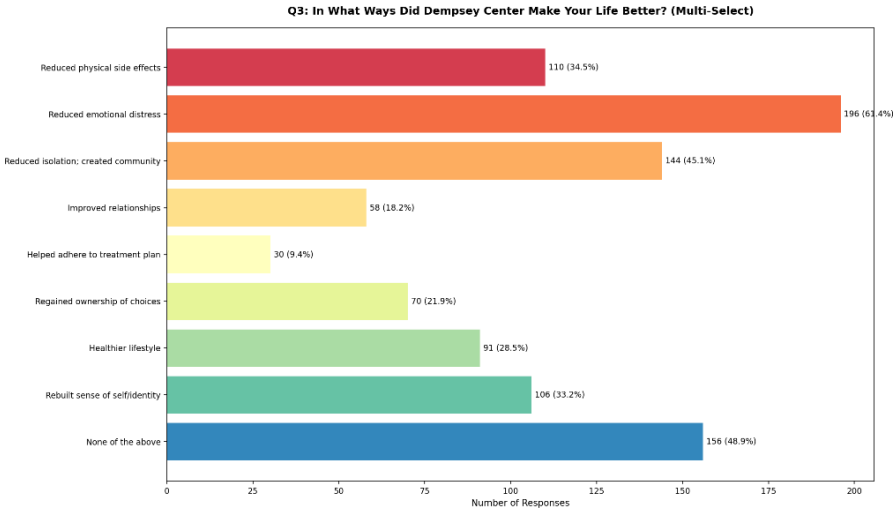

Figure 8. How Clients' Lives Improved(2025)

Service Utilization

The service frequency distribution remained broadly consistent year over year, with high-frequency usage (more than 10 times) dominating each year. 44.4% of respondents reported using services more than 10 times, while only 7% visited just once. This pattern suggests the center successfully

retains clients beyond an initial visit, and that engagement levels have not shifted significantly despite modest growth in total respondents.

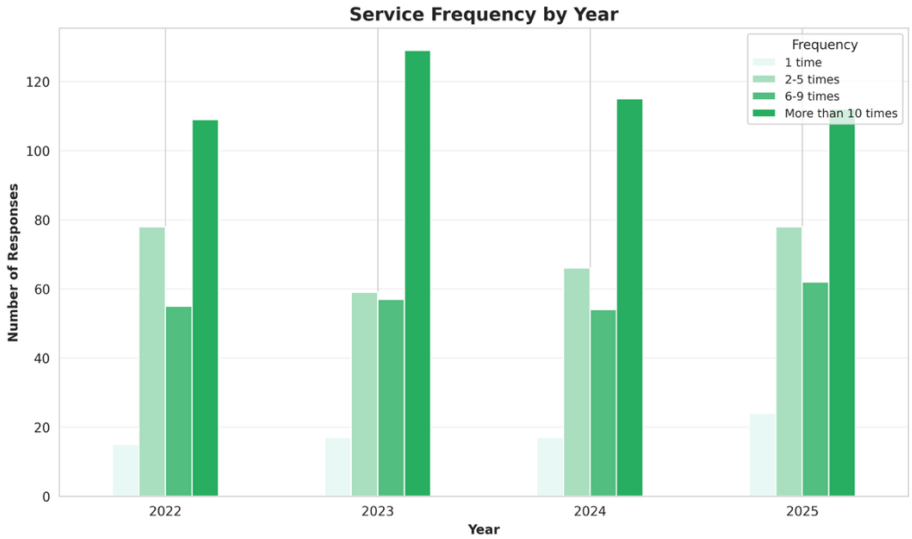

Figure 9. Service frequency distribution by year.

Westbrook (Rock Row) emerges as the most utilized location, drawing over half of all clients. Virtual live services rank a close second at nearly 46%, demonstrating strong demand for remote care options. The data reveals a clear preference for hybrid access, combining in-person convenience with digital flexibility.

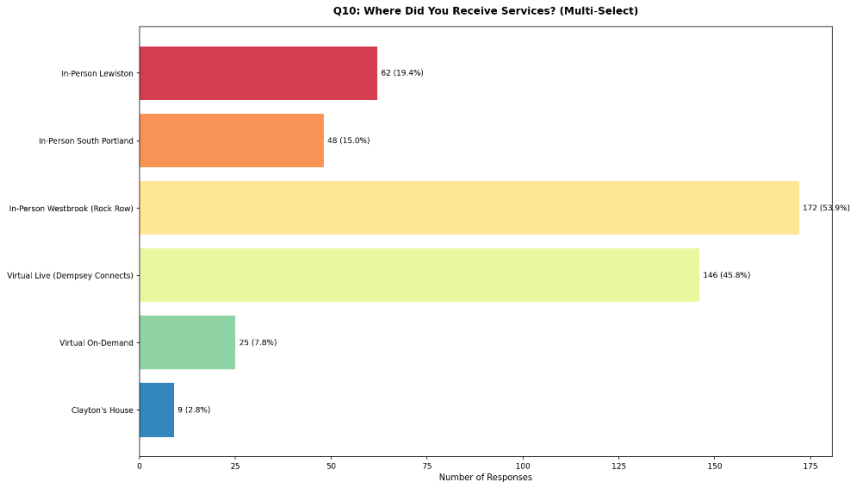

Figure 10. Service Location Preferences(2025)

B. Statistical Tests

Three chi-square tests of independence were conducted to assess whether demographic and behavioral variables were associated with high satisfaction (defined as reporting "Very Satisfied"). Service frequency showed the strongest and most significant association with satisfaction (χ^2=66.05, df=3, p<0.0001). The relationship follows a clear dose-response pattern: clients who visited more

than 10 times reported high satisfaction at a rate of 86%, compared to only 49% among those who visited just once.

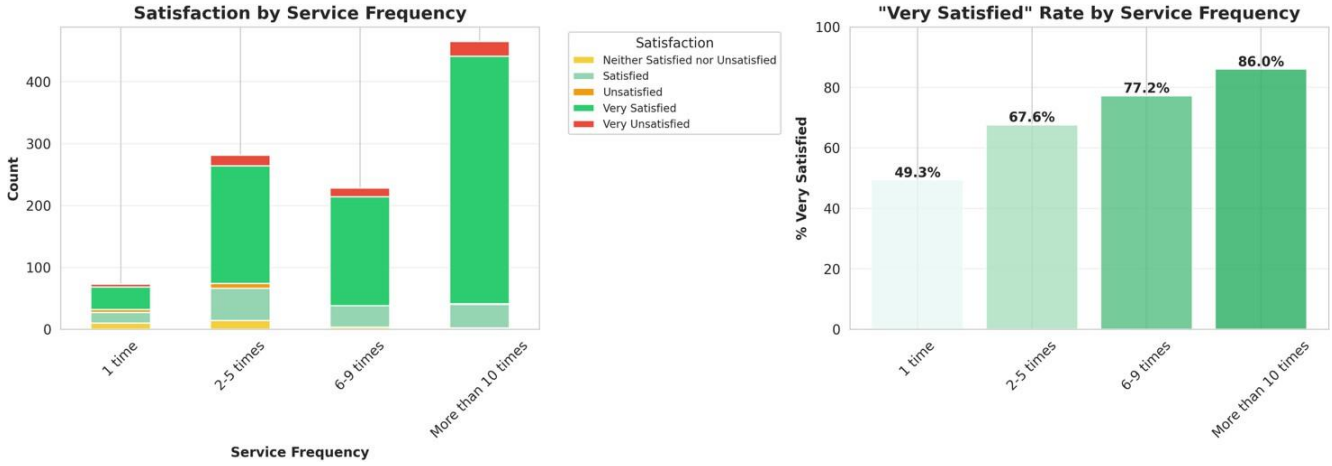

Figure 11. High satisfaction rate by service frequency.

Age was also significantly associated with satisfaction (χ^2=13.18, df=5, p=0.022). Clients aged 75 or older had the highest satisfaction rate at 84.7%, while the 25–34 group had the lowest at 62.5%. The general trend suggests older clients tend to report higher satisfaction, though sample sizes for younger age groups were small.

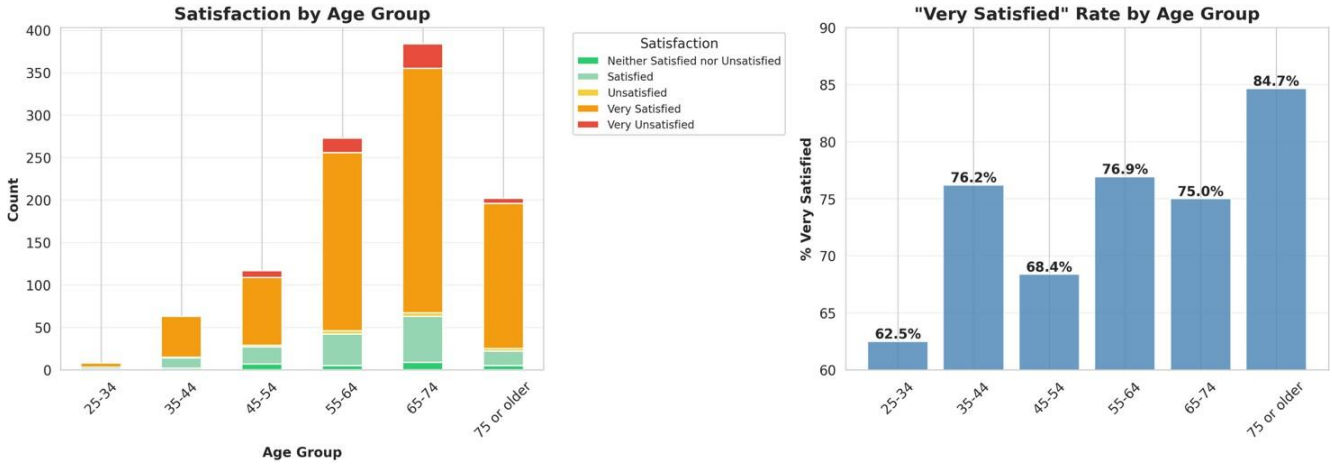

Figure 12. High satisfaction rate by age group.

Cancer status was not significantly associated with satisfaction (χ^2=7.73, df=5, p=0.172). Satisfaction rates were consistent across groups, ranging from 76.5% among current patients to 79.3% among survivors, suggesting that where a client is in their cancer journey does not meaningfully predict their satisfaction with Dempsey Center services.

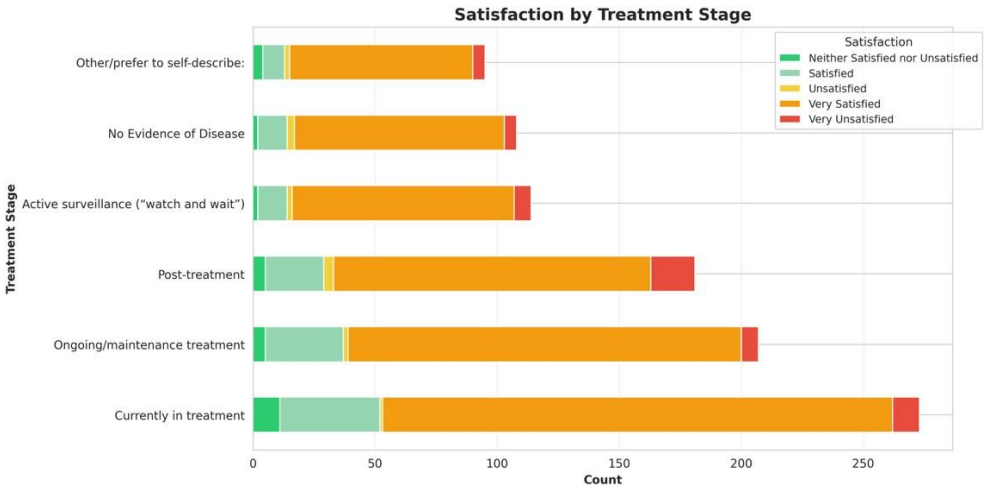

Figure 13. Treatment stage distribution.

Table 2 summarizes all three results.

Variable	χ^2	df	p-value	Significant?
Service Frequency	66.05	3	<0.0001	Yes
Age	13.18	5	0.022	Yes
Cancer Status	7.73	5	0.172	No

Table 2. Summary of chi-square test results.

C. Machine Learning Classification

We trained four classifiers to predict whether a client would report "Very Satisfied," using age, gender, service frequency, and cancer status as input features. Table 3 shows how each model performed on the held-out test set and under 5-fold cross-validation.

Model	Test Accuracy	CV Accuracy (±SD)
Gaussian Naive Bayes	74.6%	76.1% ± 1.4%
KNN (k=3)	67.9%	69.8% ± 2.9%
Decision Tree	72.1%	75.4% ± 1.3%
Random Forest	72.1%	75.6% ± 0.8%

Table 3. Model performance on test set and 5-fold cross-validation.

All four models performed in a fairly narrow range, which isn't surprising given that we only have four input features. Gaussian Naive Bayes ended up being the most reliable overall — it had the highest test accuracy (74.6%) and the highest CV accuracy (76.1%) with the tightest standard deviation (±1.4%), meaning it generalized consistently across folds rather than just getting lucky on one split. Decision Tree and Random Forest tied on test accuracy at 72.1%, though Random Forest had a slightly more stable CV performance (±0.8% vs ±1.3%). KNN lagged behind on both metrics and had the widest variance across folds (±2.9%), which suggests it struggled with the relatively small feature space.

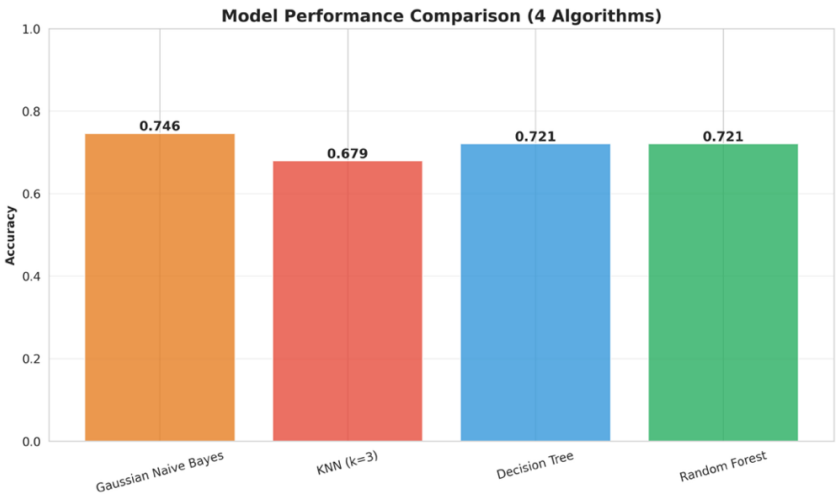

Figure 14. Model accuracy comparison across all four algorithms.

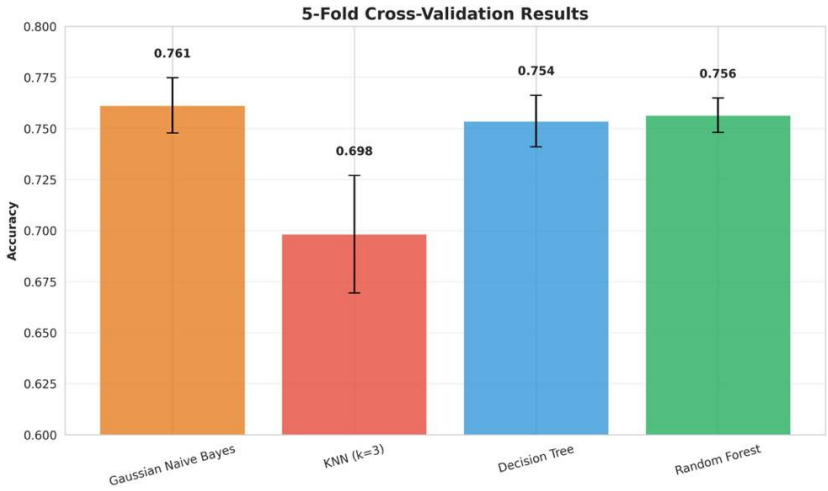

Figure 15. 5-fold cross-validation accuracy for each classifier.

D. Feature Importance Analysis

Both tree-based models ranked service frequency as the most important predictor by a significant margin — the Decision Tree gave it an importance score of 0.446, and Random Forest gave it 0.413. Age and cancer status each contributed at a secondary level — around 0.22–0.27 across both models — while gender was consistently the least important feature, scoring only 0.085–0.106.

Figure 16. Decision Tree feature importance.

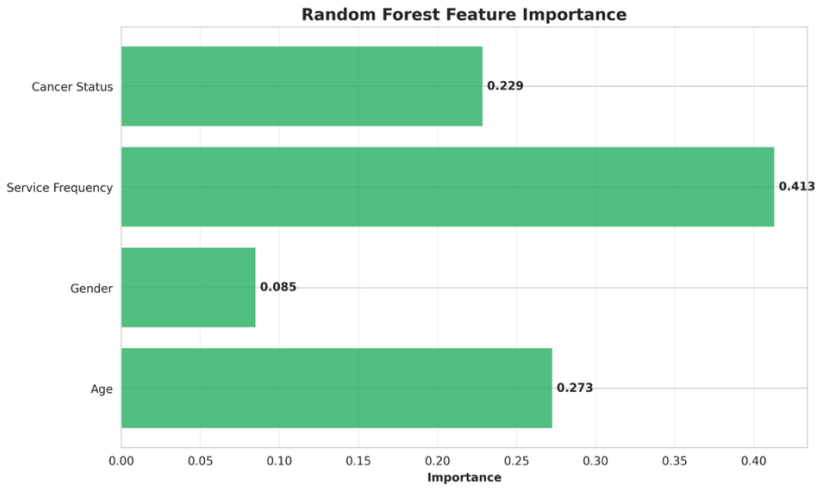

Figure 17. Random Forest feature importance.

This result lines up with everything else we found. The chi-square tests showed service frequency had the strongest association with satisfaction, the correlation analysis confirmed it had the highest r value, and now both tree models are independently arriving at the same conclusion. Gender being near the bottom also makes sense — with 81% of respondents being women, there just isn't enough variation in that variable for the models to learn much from it.

The dominance of service frequency over demographic variables like age and cancer status is worth highlighting for the Dempsey Center. It suggests that how often a client engages with services matters more than who they are. That's actually an actionable finding — the center can't change a client's age or cancer status, but it can design programs and outreach strategies that encourage people to come back more than once.

E. Confusion Matrices

Both confusion matrices show strong performance on the "Very Satisfied" class but very limited recall on the minority classes. The Decision Tree correctly classified 220 out of 232 Very Satisfied

cases, but only identified 7 out of 83 non-Very-Satisfied cases. Random Forest performed similarly, correctly classifying 224 Very Satisfied cases but catching only 3 of the 83 non-satisfied ones. This is expected — with 76% of responses falling in the "Very Satisfied" category, the models naturally lean toward predicting the majority class. In practice, both models are better at identifying satisfied clients than flagging dissatisfied ones, which is a known limitation of working with imbalanced datasets.

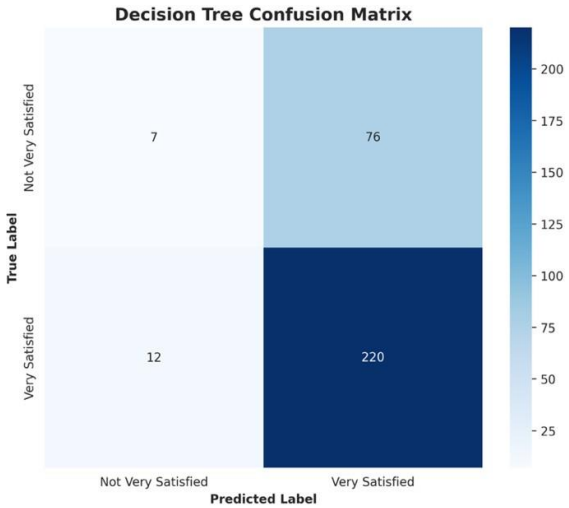

Figure 18. Decision Tree confusion matrix.

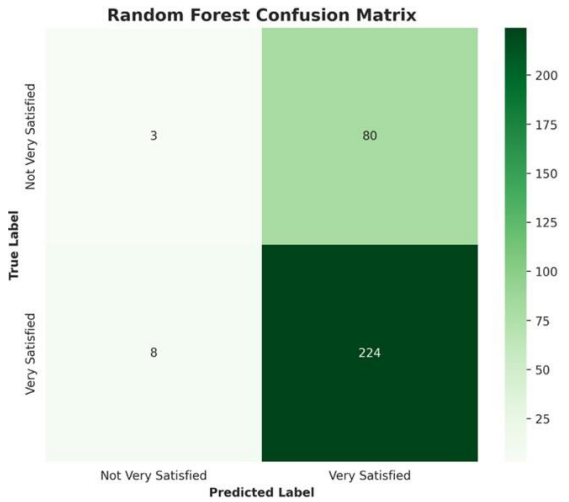

Figure 19. Random Forest confusion matrix.

F. **Correlation Analysis**

The Pearson correlation heatmap confirmed that service frequency had the strongest linear relationship with high satisfaction (r=0.247). This is modest in absolute terms but consistent and meaningful given that we're working with categorical variables encoded as integers. Age showed a weak positive correlation (r=0.079), and gender and cancer status were essentially flat at r=−0.017

and r=−0.033 respectively — consistent with the chi-square finding that those two variables aren't meaningfully associated with satisfaction outcomes.

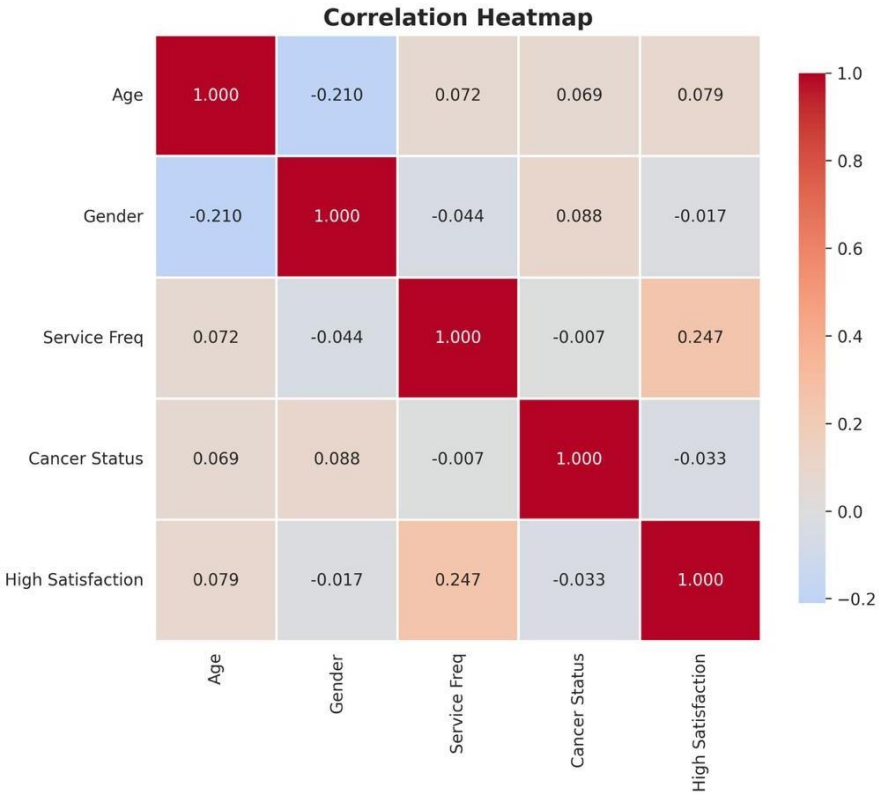

Figure 20. Pearson correlation heatmap.

V. Discussion

The clearest takeaway from this analysis is that service frequency is the single most important factor associated with client satisfaction, and it showed up that way across every method we used — chi-square tests, feature importance from both tree models, and correlation analysis all pointed to the same conclusion. Clients who visited more than 10 times were satisfied at nearly double the rate of one-time visitors. Whether that's because frequent engagement builds trust and familiarity, or because already-satisfied clients simply return more often, the practical implication is the same: getting clients to come back beyond their first visit matters a lot.

Age was the second most consistent factor. Older clients, especially those 75 and above, reported the highest satisfaction, while younger clients — particularly the 25–34 group — were the least satisfied and the most underrepresented in the data. This gap could reflect differences in what younger working-age adults need from a support service, or simply that the current program mix is better calibrated for older populations. Either way, it's worth paying attention to.

To investigate why clients with fewer than ten visits and those under age 65 reported comparatively lower satisfaction scores, we conducted a stratified barrier analysis. We divided each group internally into low-satisfaction and high-satisfaction subgroups and compared their self-reported obstacles. This within-group contrast isolates the specific access barriers that distinguish dissatisfied clients from satisfied clients within the same demographic profile.

Respondents in the low-satisfaction subgroup were substantially more likely to report that service schedules did not fit their availability (+18.1 percentage points), that they faced long waits for needed services (+11.2 pp), and that they experienced long waits to become a client (+9.9 pp). In contrast, highly satisfied low-visit users were far more likely to report encountering no barriers at all (+14.7 pp)

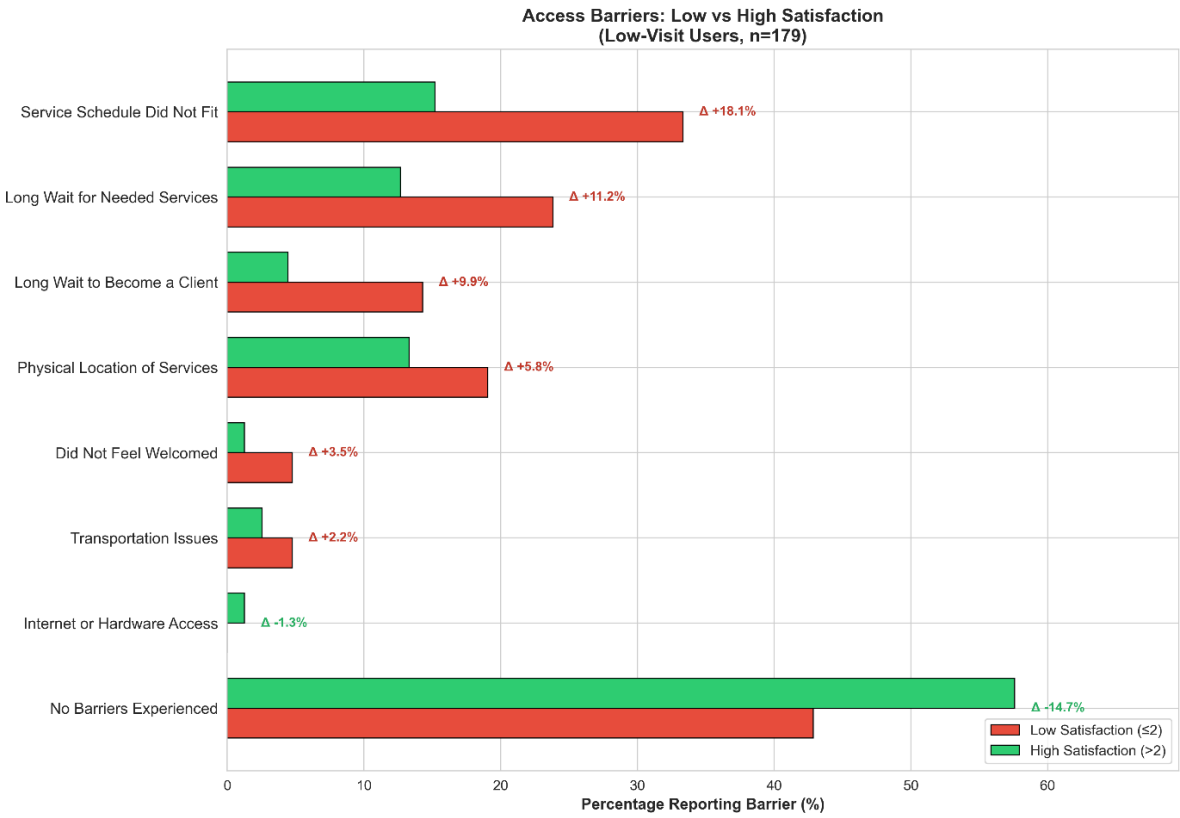

Figure 21. Access barriers reported by low-visit users (<10 visits)

Comparing low- and high-satisfaction subgroups, the divergence is even more pronounced in this cohort: low-satisfaction individuals were significantly more likely to cite mismatched service schedules (+19.3 pp) and long waits for services (+16.8 pp) than their highly satisfied peers. This pattern suggests that for younger clients, scheduling flexibility and timeliness of care is a critical driver of satisfaction.

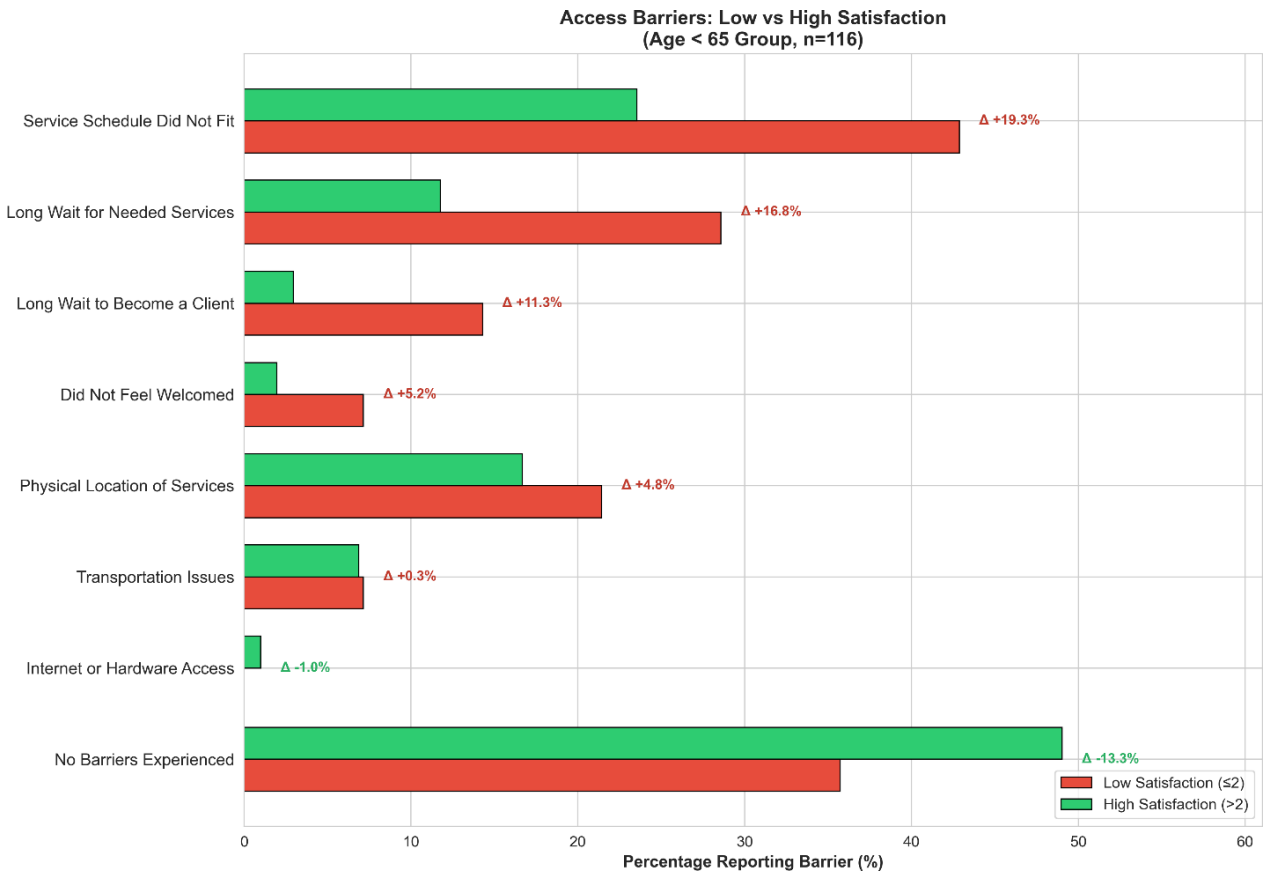

Figure 22. Access barriers reported by respondents under age 65

VI. Limitations

A few limitations are worth keeping in mind. First, the survey only captures clients who actively completed the annual questionnaire. People who had negative experiences or stopped using services altogether are unlikely to show up in the data, which probably inflates the overall satisfaction numbers.

Second, the survey instrument changed across the four years we studied, with column counts growing from 72 to 105. We manually verified the column positions for each variable, but some variables — like income and treatment stage — were only available for certain years and couldn't be included consistently across the full longitudinal analysis.

Third, the class imbalance in the outcome variable (76% "Very Satisfied") limited the classifiers' ability to detect dissatisfied clients. All four models performed well on the majority class but struggled with minority ones. Future work could apply resampling techniques like SMOTE to address this.

Finally, the features available to us were fairly limited — four demographic and behavioral variables. A richer feature set, including which specific services clients used or how long they had been attending, would likely improve both model accuracy and the depth of insights we could draw.

VII. Future Work

The most valuable next step would be tracking individual clients across multiple years rather than treating each survey response as independent. Right now we can see that satisfaction is stable at the population level, but we can't tell whether individual clients' experiences are improving, declining, or staying the same over time. A longitudinal client-level dataset would answer that question directly.

It would also be worth investing in outreach to low-engagement clients — particularly those who visited only once or twice — to understand why they didn't return after one or two visits. Exit surveys or targeted follow-ups could surface barriers that the annual satisfaction survey isn't designed to capture.

On the analysis side, applying NLP to open-ended survey responses could reveal themes and pain points that structured questions miss. And as noted above, resampling methods like SMOTE could help the classification models perform better on dissatisfied clients, who are arguably the most important group to identify correctly.

References

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